CERTIFIED ETHICAL HACKER | SCANNING NETWORKS

Network Scanning

Concepts, Tools, Techniques, IDS/Firewall Evasion, Banner Grabbing, Network Diagrams and Labs

Section	Topics covered
1	Network scanning concepts
2	Scanning tools
3	Scanning techniques
4	Scanning beyond IDS and firewall
5	Banner grabbing
6	Drawing network diagrams
7	Scanning countermeasures
8	Labs: host discovery (Nmap), scanning techniques (Nmap, Hping3), OS discovery (NSE), custom packets beyond IDS/firewall, scanning with Metasploit
9-10	Quick revision sheet and practice questions

Important: Scanning sends packets to live systems and can disrupt fragile devices or trigger security alerts. Use these techniques only on systems and networks you own or have **written authorization** to test (for example a CEH lab or your own virtual lab).

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1. Network Scanning Concepts

What is network scanning?

Network scanning is the process of identifying **live hosts, open ports, running services, operating systems and vulnerabilities** on a target network. It follows footprinting and comes before enumeration and exploitation. The results show the attack surface: which machines exist and which doors are open.

Types of scanning

Type	Purpose
Network (host) scanning	Finds live hosts and IP addresses in a range.
Port scanning	Finds open, closed and filtered TCP/UDP ports and the services behind them.
Vulnerability scanning	Checks identified systems and services for known weaknesses (e.g. Nessus, OpenVAS).

Objectives

- Discover live hosts, their IP addresses and open ports
- Discover operating systems and system architecture
- Discover services running on hosts and their versions
- Discover vulnerabilities in live hosts
- Gather information to draw a network map

TCP/IP communication essentials

TCP is connection-oriented. A connection is created with the **three-way handshake**: client sends SYN, server replies SYN/ACK, client replies ACK. A connection is ended with FIN/ACK exchanges or aborted with RST. Most scanning techniques work by sending unusual flag combinations and interpreting the replies.

Flag	Meaning
SYN	Synchronize: starts a connection
ACK	Acknowledges received data
FIN	Finish: gracefully ends a connection
RST	Reset: abruptly terminates or rejects a connection
PSH	Push: deliver data to the application immediately
URG	Urgent: data marked as high priority

Port ranges

Range	Name
0 - 1023	Well-known ports (HTTP 80, HTTPS 443, SSH 22, FTP 21, SMTP 25, DNS 53)
1024 - 49151	Registered ports
49152 - 65535	Dynamic / private ports

Nmap port states

State	Meaning
open	A service is actively accepting connections.
closed	Port is reachable but no service is listening (replies with RST).
filtered	A firewall or filter blocks probes; Nmap cannot tell if open or closed.
unfiltered	Reachable, but open or closed cannot be determined (seen with ACK scan).
open|filtered	No response; the port is open or filtered (UDP, FIN, NULL, Xmas scans).
closed|filtered	Cannot decide between closed and filtered (Idle scan).

2. Scanning Tools

Tool	Use
Nmap / Zenmap	The standard scanner: host discovery, port scanning, service and OS detection, NSE scripting. Zenmap is the GUI.
Hping3	Command-line packet crafter and analyzer for TCP, UDP, ICMP and raw IP; firewall testing and custom scans.
Metasploit Framework	Penetration-testing framework; auxiliary scanner modules and database integration with Nmap.
Masscan	Very fast Internet-scale port scanner.
NetScanTools Pro	Windows suite of network discovery and diagnostic tools.
Angry IP Scanner / Advanced IP Scanner	Simple, fast GUI IP and port scanners.
Unicornscan	Asynchronous, high-speed scanner.
SolarWinds Network Topology Mapper	Automatic discovery and network diagrams.
Nessus / OpenVAS	Vulnerability scanners.
Mobile tools	Fing, Network Scanner, IP Scanner on Android and iOS.

Nmap command structure

```
nmap [scan type] [options] [target]
Example: nmap -sS -p 1-1000 -T4 -oN result.txt 10.10.1.0/24
```

Option	Meaning
-p 80,443 / -p- / --top-ports 100	Choose ports / all 65535 ports / most common ports
-sV	Detect service versions
-O	Detect operating system
-A	Aggressive: -sV, -O, default scripts and traceroute
-sC / --script	Run default / specified NSE scripts
-Pn	Skip host discovery; treat all hosts as up
-n / -R	Never / always resolve DNS

Option	Meaning
-T0 to -T5	Timing templates: paranoid, sneaky, polite, normal, aggressive, insane
-v / -vv	Increase verbosity
-oN / -oX / -oG / -oA	Output as normal / XML / grepable / all formats
-iL file	Read targets from a file
--open	Show only open ports

3. Scanning Techniques

3.1 Host discovery (ping sweeps)

Host discovery checks which IP addresses are alive before port scanning, reducing time and noise.

Technique	Nmap option	How it works
ARP ping	-PR	ARP request on the local subnet; very reliable because ARP cannot be filtered locally.
ICMP echo ping	-PE	Classic ping; often blocked.
ICMP timestamp ping	-PP	Sends timestamp request; may pass when echo is blocked.
ICMP address mask ping	-PM	Sends address mask request.
TCP SYN ping	-PS[ports]	Sends SYN; any SYN/ACK or RST means host is up.
TCP ACK ping	-PA[ports]	Sends ACK; a RST means host is up. Can pass stateless filters.
UDP ping	-PU[ports]	Sends UDP to a likely closed port; ICMP port unreachable means host is up.
IP protocol ping	-PO[protocols]	Sends packets with different IP protocol numbers.
List scan / No port scan	-sL / -sn	List targets only / host discovery only (no ports).

3.2 Port scanning techniques

Scan	Option	Packet sent	Open port	Closed port
TCP connect	-sT	SYN, full handshake	Handshake completes	RST
SYN (half-open / stealth)	-sS	SYN	SYN/ACK (Nmap then sends RST)	RST
Xmas	-sX	FIN + PSH + URG	No response (open|filtered)	RST
FIN	-sF	FIN	No response (open|filtered)	RST
NULL	-sN	No flags	No response (open|filtered)	RST
ACK	-sA	ACK	RST (unfiltered)	RST (unfiltered)
Window	-sW	ACK	RST with non-zero window	RST with zero window
Maimon	-sM	FIN + ACK	No response (some BSD)	RST
UDP	-sU	UDP datagram	Reply or no reply (open|filtered)	ICMP port unreachable (type 3, code 3)

Scan	Option	Packet sent	Open port	Closed port
Idle (zombie)	-sI zombie	Spoofed SYN via zombie	Zombie IPID increments by 2	IPID increments by 1
SCTP INIT / COOKIE-ECHO	-sY / -sZ	SCTP chunks	INIT-ACK / no response	ABORT / ICMP error

Key points

- **SYN scan** never completes the handshake, so it is faster and was historically less likely to be logged by applications. It needs raw-socket (root) privileges.
- **Connect scan** uses the OS connect() call, so it works without privileges but is logged by the target.
- **Xmas, FIN and NULL** scans follow RFC 793 behaviour: closed ports send RST, open ports stay silent. Windows (and some others) send RST regardless, so these scans are unreliable against Windows.
- **ACK scan** does not find open ports. It maps **firewall rules**: RST means unfiltered, no reply or ICMP error means filtered (stateful firewall).
- **Idle scan** hides the scanner's IP by bouncing probes off a zombie host with predictable IP ID values.
- **UDP scans** are slow because closed ports rate-limit ICMP errors.

3.3 Service version and OS detection

Service detection (-sV) probes open ports and matches the replies against a signature database to report product and version. **OS detection** (-O) fingerprints the TCP/IP stack: initial TTL, window size, DF bit, TCP options and responses to special probes.

OS family	Typical initial TTL	Typical TCP window
Linux / Unix	64	5840, 29200 and others
Windows	128	8192, 64240 and others
Cisco IOS / Solaris	255	4128 / 8760

Active OS fingerprinting sends probes (Nmap, Xprobe2). **Passive** fingerprinting only observes existing traffic (p0f, Wireshark) and is stealthier. Remember that observed TTL is the initial value minus the number of hops.

4. Scanning Beyond IDS and Firewall

Firewalls filter packets by rules; Intrusion Detection Systems (IDS) alert on suspicious patterns. Testers use evasion techniques to check whether a network's defences notice or block scans. Each technique below is a **test of the defence**: a well-configured IDS/firewall should detect or block most of them.

Technique	Idea	Nmap / Hping3 example
Packet fragmentation	Split probe headers into tiny fragments so simple filters cannot match them.	nmap -f ; nmap --mtu 16
Source port manipulation	Use a port that is commonly allowed (53, 80, 443) as the source port.	nmap -g 53 ; nmap --source-port 53
IP address decoys	Mix the real source among fake sources so the target logs many addresses.	nmap -D RND:10 ; -D decoy1,ME,decoy2
IP spoofing	Forge source IP (only useful where replies are not needed or can be observed).	nmap -S spoofed -e eth0 -Pn ; hping3 -a

Technique	Idea	Nmap / Hping3 example
MAC spoofing	Change the source MAC address.	nmap --spoof-mac 0 (random)
Randomise targets / timing	Avoid sequential patterns and threshold rules.	nmap --randomize-hosts -T1 ; --scan-delay 5s
Append random data	Change packet size to avoid fixed-length signatures.	nmap --data-length 25
Bad checksum	Detect firewalls/IDS that respond to packets real hosts drop.	nmap --badsum
Proxies and anonymizers	Route the scan through intermediate hosts to hide origin.	nmap --proxies ; proxychains nmap -sT
Source routing	Specify the path for the packet, bypassing some filters (largely blocked today).	hping3 --lsrr / --ssrr (not in Nmap)
Custom flags and packets	Craft unusual flag combinations or protocols.	hping3 -F -P -U ; nmap --scanflags URGACKPSH
Idle scan	Bounce probes through a zombie host.	nmap -sI zombie:port target

Detecting a firewall

- **ACK scan** (-sA): compare filtered vs unfiltered ports to infer rules.
- **No response** to SYN: probe dropped (filtered) vs **RST** (closed) vs **ICMP unreachable type 3 code 13** (administratively prohibited).
- **TTL and banner comparison** across ports can expose a firewall or proxy answering for hosts behind it.
- **Traceroute** (--traceroute or hping3 -t) shows where probes stop.

Do not assume evasion succeeded because you got no alert; check the IDS/firewall logs afterwards and report what was and was not detected.

5. Banner Grabbing

A **banner** is the text a service returns when a client connects: software name, version, sometimes OS. Banner grabbing collects it to identify software and look up known vulnerabilities.

Type	Description
Active	Connect directly and read or provoke the response (telnet, netcat, curl, Nmap).
Passive	Observe existing traffic, error pages, search engines (Shodan), or captured packets, without touching the target.

```
telnet 10.10.1.22 21                # FTP banner
nc -nv 10.10.1.22 80                # then type: HEAD / HTTP/1.0 (press Enter twice)
curl -I http://10.10.1.22           # HTTP response headers (Server: ...)
nmap -sV --script banner -p 21,22,25,80 10.10.1.22
nmap -sV --version-intensity 5 10.10.1.22
```

Other tools: ID Serve, Netcraft, Shodan, Censys, Wget, Wireshark (passive). Useful HTTP headers: Server, X-Powered-By, X-AspNet-Version.

Banner grabbing countermeasures

- Turn off or change banners (e.g. ServerTokens Prod in Apache, remove X-Powered-By headers, IIS URLScan/Request filtering).
- Remove version info from error pages; restrict unnecessary services.
- Use a reverse proxy or WAF to normalise responses; keep software patched.

6. Drawing Network Diagrams

After discovery, scan results are turned into a **network diagram** (topology map): routers, switches, firewalls, servers, workstations and the links between them. Diagrams help to spot critical assets, segmentation gaps and attack paths, and they document the engagement.

Process

- Discover hosts and ports (Nmap, ping sweeps) and record IPs, OS and services.
- Trace routes (traceroute / tracert, Nmap --traceroute) to find routers and hops.
- Identify device roles: gateway, firewall, DMZ servers, internal servers, endpoints.
- Draw nodes and links, group by subnet/VLAN, label IP addresses and services.
- Add findings (open ports, vulnerabilities) and update after each scan.

Tool	Notes
SolarWinds Network Topology Mapper	Automated discovery with scheduled map refresh.
Network Notepad	Free Windows tool for manual diagrams.
Lucidchart, draw.io (diagrams.net), Cacoo	Online diagramming.
OpManager, LANState, Network Topology Mapper	Network monitoring with topology maps.
Zenmap topology tab	Visual map of Nmap scan results.

7. Scanning Countermeasures

- Configure firewalls to block unnecessary ports, and filter inbound ICMP (types such as timestamp and address mask).
- Use stateful inspection and ingress/egress filtering; drop packets with illegal flag combinations (e.g. NULL, Xmas) and spoofed or source-routed packets (disable source routing).
- Deploy IDS/IPS with scan-detection rules (SYN rate, port sweep thresholds, fragment reassembly).
- Close unused ports and disable unneeded services; harden hosts and keep systems patched.
- Change or remove service banners; use port-knocking or VPN for management services.
- Keep router/firewall firmware up to date; use network segmentation and monitoring; run your own regular scans to find what attackers will see.

8. Hands-on Labs

Commands below use example addresses (attacker 10.10.1.13, targets 10.10.1.0/24, Windows 10.10.1.22, Linux 10.10.1.9). Replace them with your lab machines. Run Nmap, Hping3 and Metasploit commands with **sudo** or as root where needed (Kali/Parrot). Option behaviour can vary slightly by version; use *--help* or *man* to confirm.

Lab 1: Host discovery using Nmap

Objective: find live hosts in 10.10.1.0/24 using several discovery methods.

```
nmap -sn -PR 10.10.1.0/24                # ARP ping (local network)
nmap -sn -PE 10.10.1.0/24                # ICMP echo ping sweep
nmap -sn -PP 10.10.1.0/24                # ICMP timestamp ping
nmap -sn -PM 10.10.1.0/24                # ICMP address mask ping
nmap -sn -PS 10.10.1.0/24                # TCP SYN ping (default port 80)
nmap -sn -PA 10.10.1.0/24                # TCP ACK ping
nmap -sn -PU 10.10.1.0/24                # UDP ping
nmap -sn -PO 10.10.1.0/24                # IP protocol ping
nmap -sn -PS22,80,443 10.10.1.22         # SYN ping to chosen ports
nmap -sn -n 10.10.1.0/24 -oN live.txt    # no DNS, save output
```

- Hosts reported as **Host is up** are live; note their MAC address and vendor, which hints at device type.
- Compare the results of each method. A host that ignores ICMP may answer ARP or TCP probes.
- Use **-Pn** when hosts block all discovery probes, but expect a much slower scan.

Lab 2: Network scanning techniques using Nmap and Hping3

Part A: Nmap port scans against the Windows target 10.10.1.22.

```
nmap -sT -p 1-1024 10.10.1.22            # TCP connect scan
nmap -sS 10.10.1.22                      # SYN (stealth) scan
nmap -sX -p 21,22,80,445 10.10.1.22      # Xmas scan
nmap -sF -p 21,22,80,445 10.10.1.22      # FIN scan
nmap -sN -p 21,22,80,445 10.10.1.22      # NULL scan
nmap -sA -p 80,443 10.10.1.22            # ACK scan (firewall rule mapping)
nmap -sW -p 80,443 10.10.1.22            # Window scan
nmap -sU -p 53,67,123,161 10.10.1.22     # UDP scan
nmap -sI zombie_ip:80 10.10.1.22         # Idle scan (needs a suitable zombie)
nmap -sV -p- 10.10.1.22                  # service versions on all ports
nmap -A -T4 10.10.1.22                   # aggressive scan
```

Expect Xmas/FIN/NULL to show **open|filtered** on Linux hosts but misleading results (all closed) on Windows. Use Wireshark to watch the packets and confirm the flags sent.

Part B: Hping3 probes (-c 5 sends 5 packets; -p sets destination port).

```
hping3 10.10.1.22 -1 -c 4                # ICMP ping
hping3 -S 10.10.1.22 -p 80 -c 5          # SYN: SA = open, RA = closed
hping3 -A 10.10.1.22 -p 80 -c 5          # ACK scan: RST = unfiltered
hping3 -F 10.10.1.22 -p 80 -c 5          # FIN
hping3 -F -P -U 10.10.1.22 -p 80 -c 5    # Xmas (FIN+PSH+URG)
hping3 10.10.1.22 -p 80 -c 5             # no flags set (NULL probe)
hping3 --udp 10.10.1.22 -p 53 -c 5       # UDP probe
hping3 -8 1-1024 -S 10.10.1.22           # SYN scan mode across ports
```

```
hping3 --traceroute -V -1 10.10.1.22          # route to target
```

Hping3 reply flags	Meaning
SA	SYN/ACK: port is open
RA	RST/ACK: port is closed
R	RST only: unfiltered (ACK probe)
No reply / ICMP error	Filtered or host down

Note: by default hping3 sends a TCP packet with **no flags**, which makes it a NULL probe. Options -S, -A, -F, -R, -P and -U set individual flags; combine them to craft custom flag patterns.

Lab 3: OS discovery using Nmap Script Engine (NSE)

Objective: identify the operating system with built-in detection and NSE scripts.

```
nmap -O 10.10.1.22                            # OS fingerprint
nmap -O --osscan-guess 10.10.1.22             # show guesses when uncertain
nmap -sV -O --osscan-limit 10.10.1.0/24       # only fingerprint promising hosts
nmap --script smb-os-discovery.nse 10.10.1.22   # OS, computer name, domain, time
nmap --script smb-os-discovery -p 445 10.10.1.22
nmap -sV --script banner,http-server-header -p 21,22,80,443 10.10.1.9
nmap -A 10.10.1.9                             # OS + versions + default scripts
nmap --script-help smb-os-discovery           # read what a script does
```

- **smb-os-discovery** returns OS version, computer name, domain/workgroup and FQDN from SMB sessions on ports 139/445.
- Read the lines **OS details**, **Aggressive OS guesses** and **Network Distance** (hops).
- Scripts are organised in categories (default, safe, discovery, vuln, intrusive); choose with --script "category".
- Cross-check with TTL observations from ping to catch wrong guesses.

Lab 4: Custom TCP and UDP packets using Nmap and Hping3 beyond IDS/firewall

Objective: test how a firewall/IDS reacts to evasive probes. Run Wireshark on the target (or Snort/Suricata logs) to observe what is seen. Use a lab firewall you control.

Part A: Nmap evasion options

```
nmap -f 10.10.1.22                            # fragment packets (8-byte)
nmap --mtu 16 10.10.1.22                      # custom fragment size (multiple of 8)
nmap -sS -g 53 -Pn 10.10.1.22                 # source port 53 (DNS)
nmap -D RND:10 10.10.1.22                     # 10 random decoys
nmap -D 10.10.1.5,10.10.1.6,ME 10.10.1.22     # chosen decoys
nmap --data-length 25 10.10.1.22              # random padding
nmap --spoof-mac 0 10.10.1.22                 # random MAC (LAN)
nmap -T1 --scan-delay 5s --randomize-hosts 10.10.1.0/24   # slow and random
nmap --badsum 10.10.1.22                      # bad checksum test
nmap --scanflags URGACKPSH -p 80 10.10.1.22   # custom flag combination
nmap -sS -Pn -f -T2 -g 53 --data-length 20 10.10.1.22     # combined
```

Part B: Hping3 custom packets

```
hping3 -S 10.10.1.22 -p 80 -s 53 -c 5          # SYN from source port 53
hping3 -S 10.10.1.22 -p 80 -f -c 5             # fragmented packets
hping3 -S 10.10.1.22 -p 80 -d 120 -c 5         # 120 bytes of data
hping3 -S 10.10.1.22 -p 80 -a 10.10.1.99 -c 3  # spoofed source (lab only)
hping3 -S 10.10.1.22 -p 80 --tcp-timestamp -c 5 # TCP timestamp option
hping3 --udp 10.10.1.22 -p 161 -d 40 -c 5      # custom UDP probe
hping3 -S 10.10.1.22 -p ++20 -i u100000 -c 10  # incrementing port, interval
hping3 --scan 20-100 -S 10.10.1.22             # scan range of ports
```

What to record: which probes were answered, which were dropped, and whether the IDS raised alerts. A defence that blocks the plain scan but not the fragmented or spoofed-source variant has a gap to report.

Lab 5: Scanning the target network using Metasploit

Objective: use the Metasploit Framework's database and auxiliary scanner modules for discovery and enumeration. The PostgreSQL database must be running to store results.

Steps

- Start the database service and Metasploit console.
- Verify the database connection, and create or switch to a workspace.
- Run **db_nmap** to scan and import results automatically.
- Review stored data with **hosts** and **services**.
- Use auxiliary scanner modules for focused scans (set RHOSTS, THREADS, then run).

```
sudo service postgresql start          # (or: sudo msfdb init)
msfconsole
msf6 > db_status
msf6 > workspace -a lab1
msf6 > db_nmap -sS -sV -O 10.10.1.0/24
msf6 > hosts
msf6 > services
msf6 > services -p 445 -u              # only up services on port 445
msf6 > use auxiliary/scanner/discovery/arp_sweep
msf6 auxiliary(arp_sweep) > set RHOSTS 10.10.1.0/24
msf6 auxiliary(arp_sweep) > run
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(tcp) > set RHOSTS 10.10.1.22
msf6 auxiliary(tcp) > set PORTS 1-1024
msf6 auxiliary(tcp) > set THREADS 10
msf6 auxiliary(tcp) > run
msf6 > use auxiliary/scanner/smb/smb_version
msf6 auxiliary(smb_version) > set RHOSTS 10.10.1.0/24
msf6 auxiliary(smb_version) > run
msf6 > use auxiliary/scanner/discovery/udp_sweep
msf6 > search portscan                 # find other scanner modules
```

Module	Purpose
auxiliary/scanner/discovery/arp_sweep	Finds live hosts on the local subnet via ARP
auxiliary/scanner/discovery/udp_sweep	Probes common UDP services
auxiliary/scanner/portscan/tcp, syn, ack, xmas	Different port scan types
auxiliary/scanner/smb/smb_version	Identifies SMB and OS version

Module	Purpose
auxiliary/scanner/http/http_version, ssh/ssh_version, ftp/ftp_version	Banner and version grabbing

Using db_nmap means Nmap output is stored in the Metasploit database, so later modules can use **hosts -R** to set RHOSTS automatically. This lab is scanning only; do not run exploit modules unless your scope covers it.

9. Quick Revision Sheet

Topic	Remember
Scanning	Finds live hosts, open ports, services, OS, vulnerabilities; after footprinting.
Three-way handshake	SYN, SYN/ACK, ACK. SYN scan sends RST after SYN/ACK (half-open).
Xmas / FIN / NULL	FIN+PSH+URG / FIN / no flags. Open = no reply; closed = RST. Not reliable on Windows.
ACK scan	Maps firewall rules: RST = unfiltered, no reply = filtered.
UDP scan	Closed = ICMP type 3 code 3; open often silent.
Idle scan	Uses a zombie and IPID; hides your IP.
Host discovery	-PR ARP, -PE echo, -PP timestamp, -PM mask, -PS SYN, -PA ACK, -PU UDP, -PO protocol, -sn no ports.
Evasion	-f, --mtu, -D, -g, --data-length, --spoof-mac, -T, --badsum, proxies.
OS detection	-O, -A, smb-os-discovery; TTL 64 Linux, 128 Windows, 255 Cisco.
Banner grabbing	telnet, nc, curl -I, nmap -sV; defend by hiding versions.
Hping3	-S SYN, -A ACK, -F FIN, -1 ICMP, --udp, -s source port, -a spoof, -c count.
Metasploit	db_nmap, hosts, services, auxiliary/scanner modules.
Countermeasures	Firewall + IDS, close ports, filter ICMP, disable source routing, hide banners.

10. Practice Questions

#	Question	Answer
1	Which Nmap option performs a TCP half-open (stealth) scan?	-sS.
2	Which scan sends FIN, PSH and URG flags?	Xmas scan (-sX).
3	What does an RST reply to an ACK scan indicate?	The port is unfiltered (not blocked by a stateful firewall).
4	What does a closed UDP port return?	ICMP port unreachable (type 3, code 3).
5	Which option disables host discovery and treats all hosts as up?	-Pn.
6	Which Nmap option fragments probe packets?	-f (or --mtu with a multiple of 8).
7	Which option makes Nmap use decoy source addresses?	-D.
8	Which Nmap option sets the source port?	-g or --source-port.
9	Which scan uses a zombie host's IPID?	Idle scan (-sI).
10	Typical initial TTL of Windows and Linux systems?	Windows 128, Linux/Unix 64.
11	Which NSE script reveals OS and computer name over SMB?	smb-os-discovery.

#	Question	Answer
12	Which Hping3 flag sets a spoofed source address?	-a (--spoof).
13	What is the purpose of banner grabbing?	To identify service software, versions and sometimes OS in order to find known vulnerabilities.
14	Which Metasploit command runs Nmap and stores results in its database?	db_nmap.
15	Name two ways to defend against banner grabbing.	Remove or alter service banners and version headers; use reverse proxies/WAF and patch software.

Practice every lab in an isolated, authorized environment and keep screenshots of commands and results for your notes.